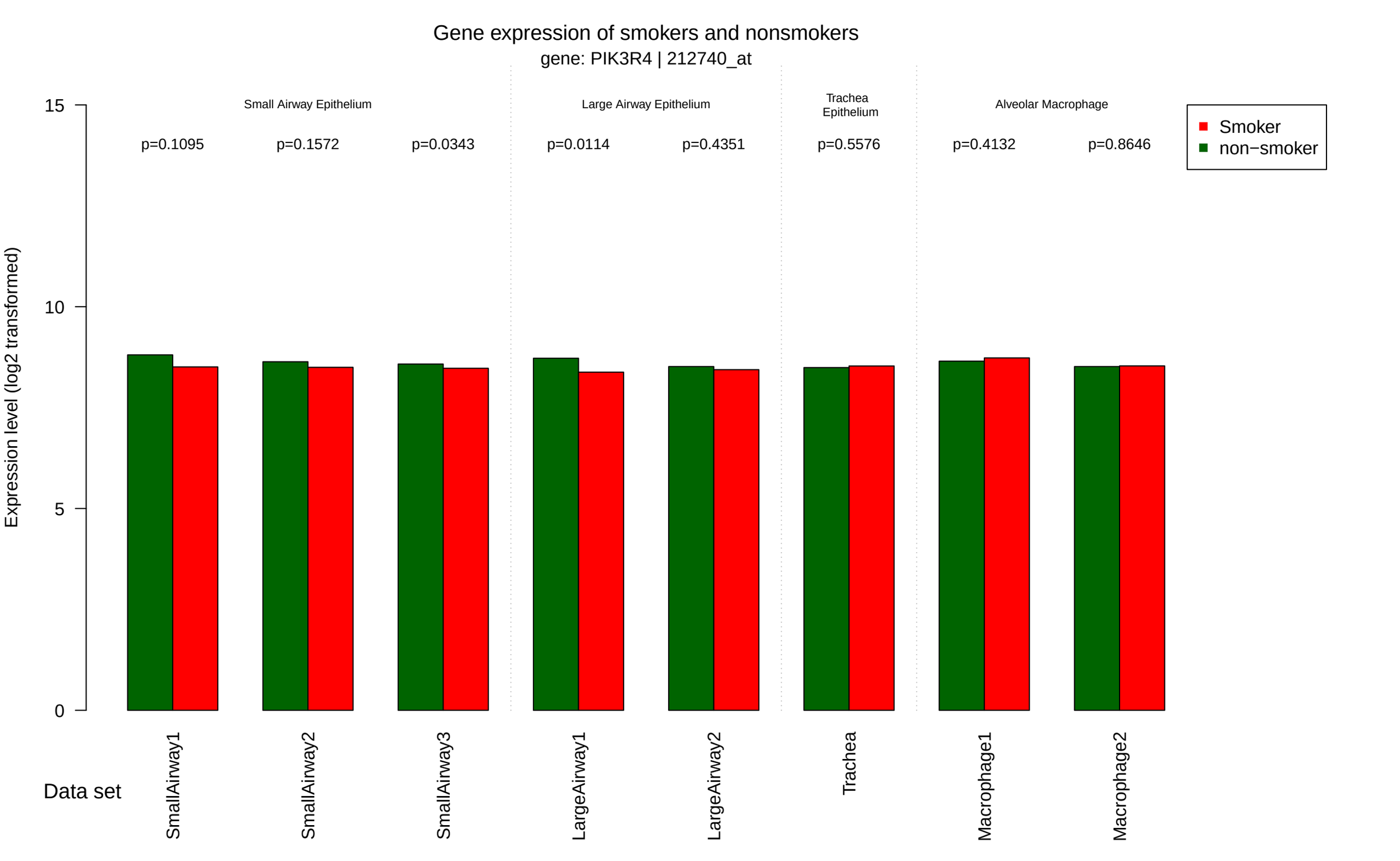
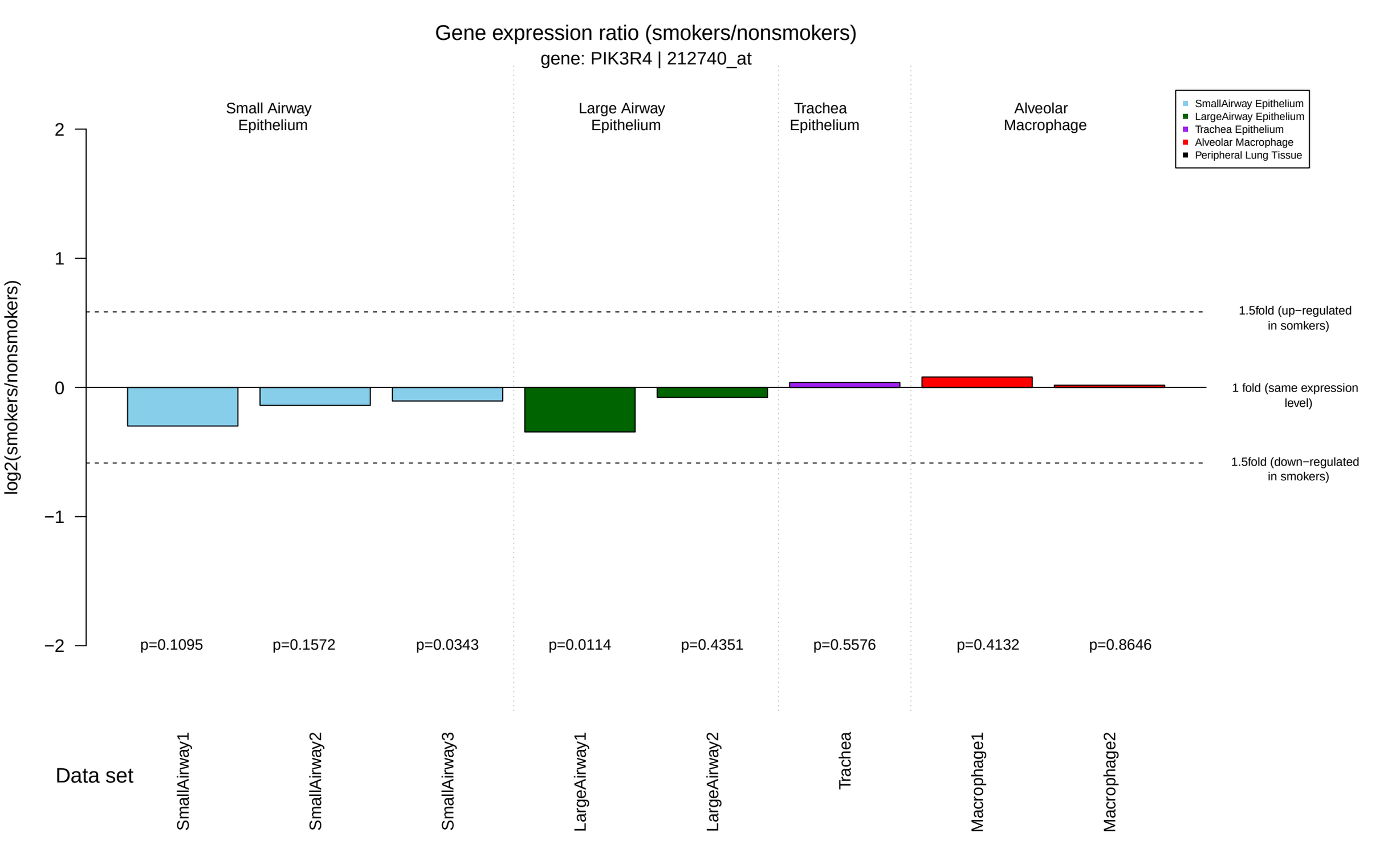
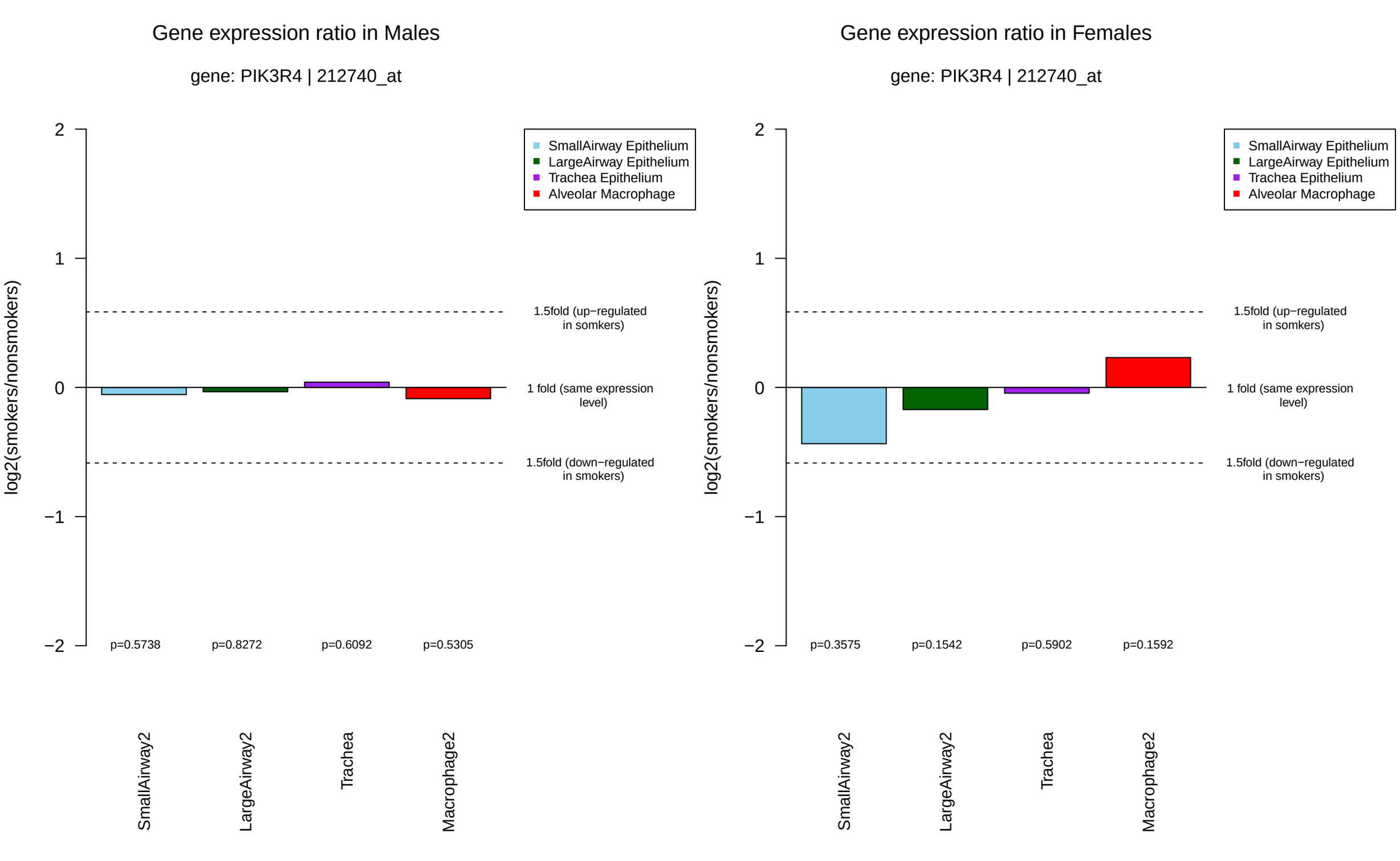
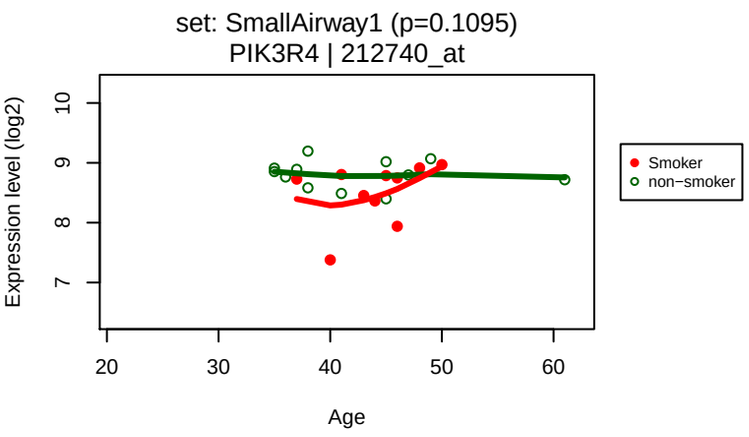
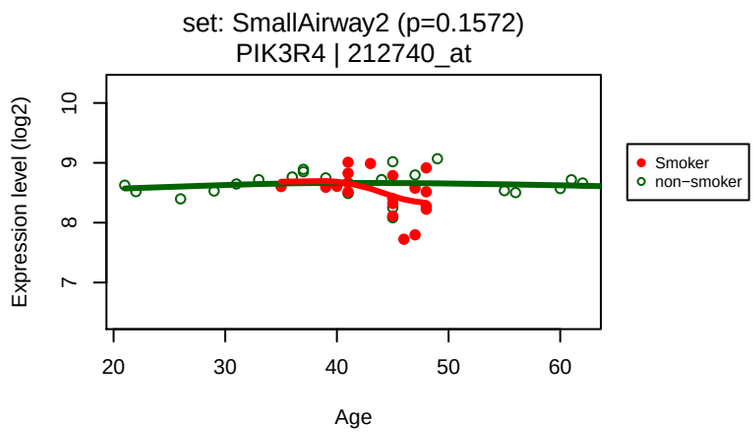
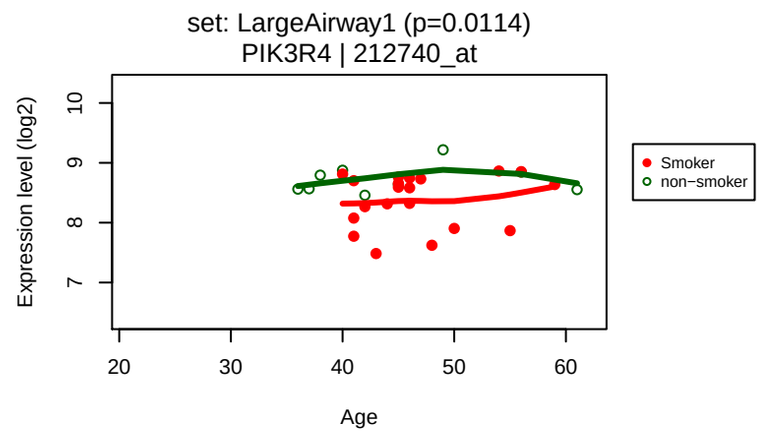

Gene expression of smokers and nonsmokers
gene: PIK3R4 | 212740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1095
p=0.1572
p=0.0343
p=0.0114
p=0.4351
p=0.5576
p=0.4132
p=0.8646
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

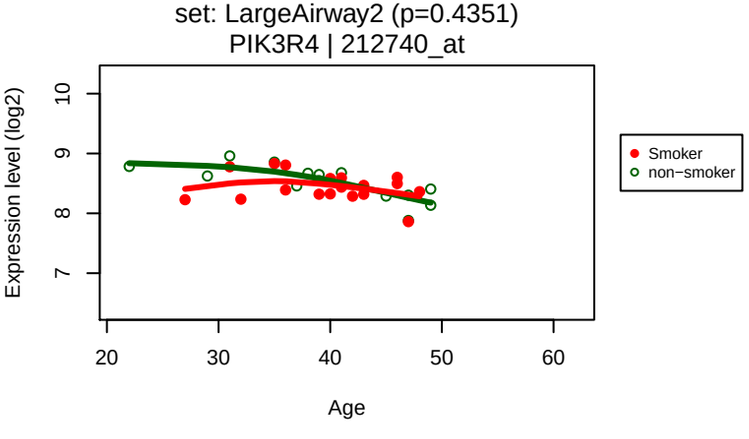

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R4 | 212740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1095
p=0.1572
p=0.0343
p=0.0114
p=0.4351
p=0.5576
p=0.4132
p=0.8646
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

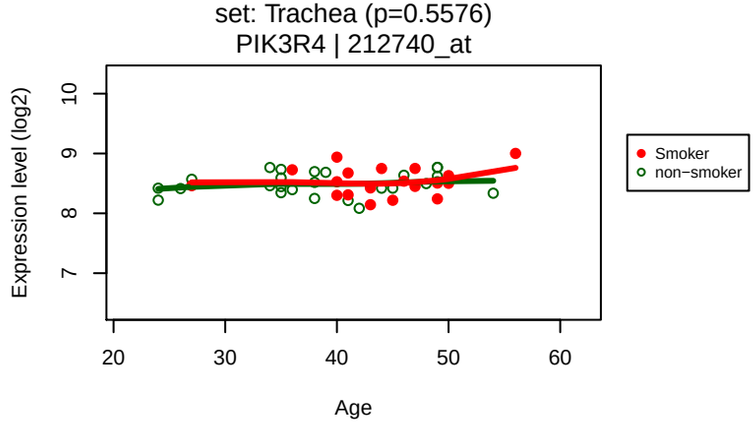

Gene expression ratio in Males
gene: PIK3R4 | 212740_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5738
p=0.8272
p=0.6092
p=0.5305
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R4 | 212740_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3575
p=0.1542
p=0.5902
p=0.1592
SmallAirway2
LargeAirway2
Trachea
Macrophage2

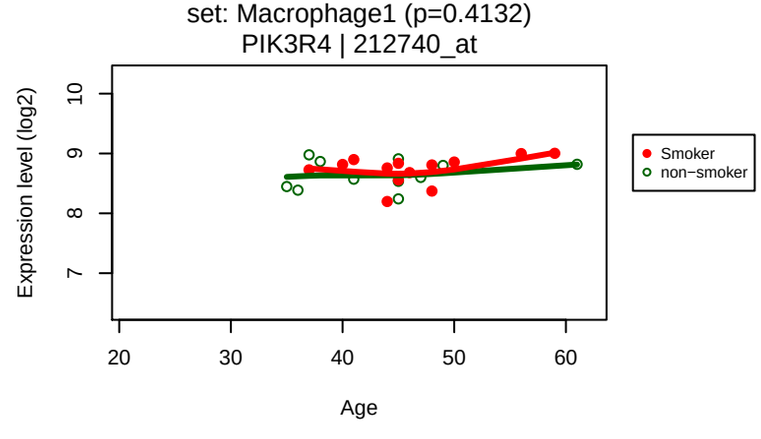

set: SmallAirway1 (p=0.1095)
PIK3R4 | 212740_at
Expression level (log2)
Age
Smoker
non−smoker

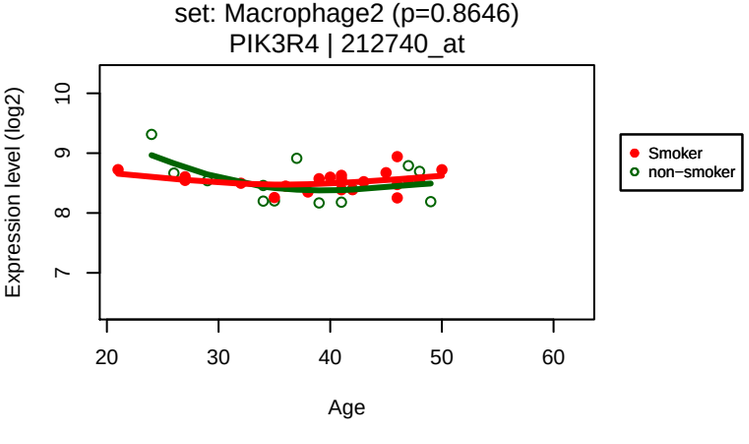

set: SmallAirway2 (p=0.1572)
PIK3R4 | 212740_at
Expression level (log2)
Age
Smoker
non−smoker

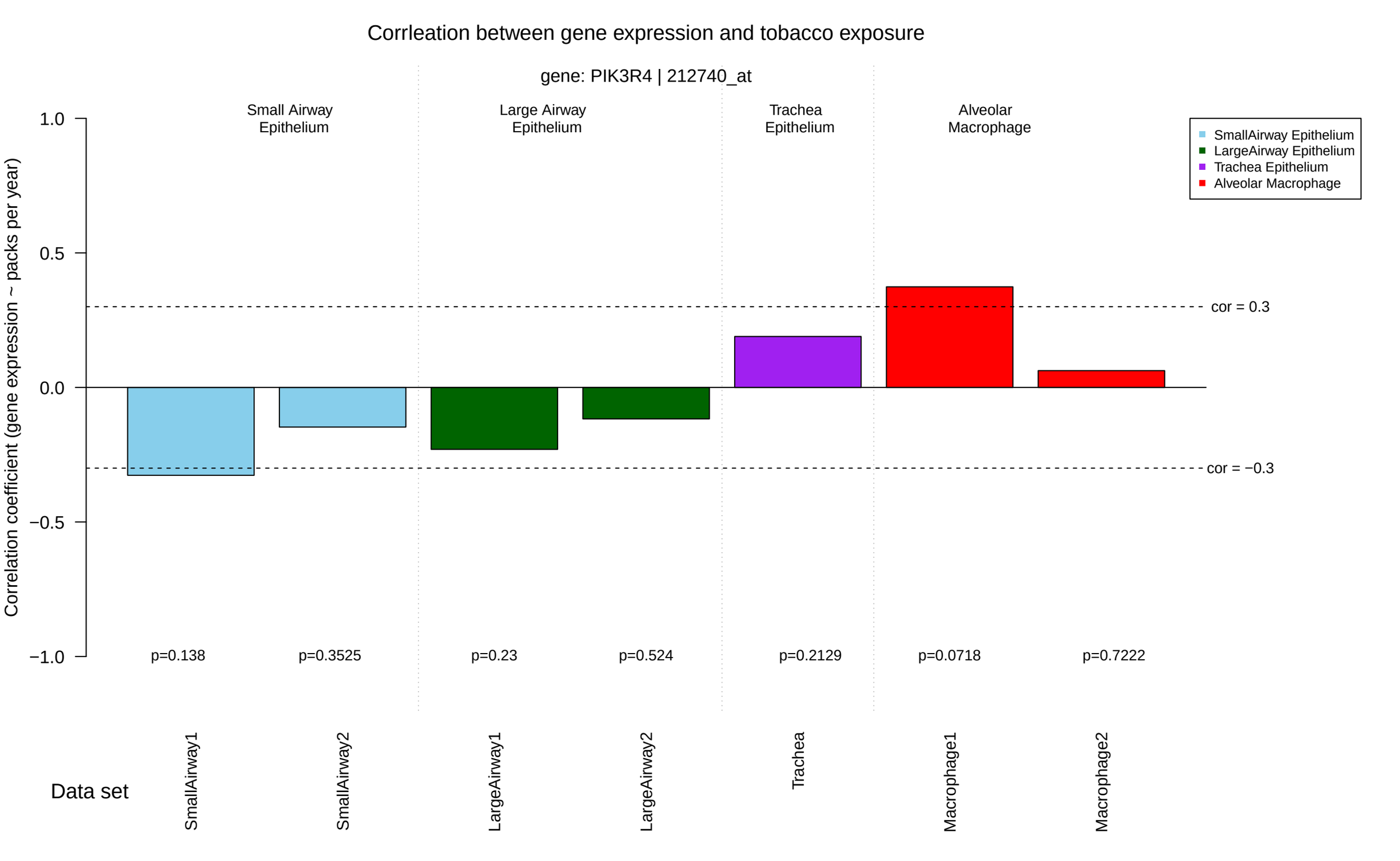

set: LargeAirway1 (p=0.0114)
PIK3R4 | 212740_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4351)
PIK3R4 | 212740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5576)
PIK3R4 | 212740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4132)
PIK3R4 | 212740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8646)
PIK3R4 | 212740_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R4 | 212740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.138
p=0.3525
p=0.23
p=0.524
p=0.2129
p=0.0718
p=0.7222
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2